

ASG AIRLINES

End-to-End Data Engineering Case Study

Detailed Technical Documentation

Project	ASG Airlines Data Engineering
Processing	Python / Pandas
Reporting	Microsoft Power BI
Documentation	Architecture, DFD, Data Model and KPIs

Contents

1 Executive Summary 3

2 Problem Statement 3

3 Project Objectives 3

4 Dataset Description 4

4.1 Flight Dataset Structure 4

5 End-to-End Solution Architecture 5

5.1 Architecture Layers 5

6 Data Flow Diagram 5

7 Data Quality Assessment 6

7.1 Duplicate Analysis 6

7.2 Flight Identifier Validation 6

8 Missing-Value Handling 7

8.1 Airline Imputation Logic 7

9 Data Standardization 7

10 Flight Duration Calculation 8

10.1 Duration Validation 8

11 Overnight Flight Handling 8

11.1 Invalid Temporal Record 9

12 Statistical Anomaly Detection 9

13 PII Protection 10

13.1 Hashing 10

13.2 Email Masking 10

13.3 Phone Masking 10

13.4 Data Minimization 11

14 Privacy and Access Control 11

15 Data Validation 11

16 Analytical Data Model	12
16.1 Relationship Design	12
17 Duplicate-Key Modelling Considerations	12
18 Business KPIs	12
19 Airline Distribution	13
20 Route-wise Traffic	13
21 Booking Status Analysis	14
22 Delay and Anomaly Analysis	15
23 Power BI Dashboard	15
23.1 KPI Cards	16
23.2 Interactive Visuals	16
23.3 Operational Interpretation	16
24 Data Quality Governance	17
25 Scalability and Performance	17
26 Error Handling and Logging	18
27 Assumptions and Business Rules	18
28 Key Design Decisions	18
28.1 Do Not Treat Every Repeated ID as a Duplicate	19
28.2 Do Not Convert Missing Payments to Zero	19
28.3 Do Not Invent Scheduled Flight Times	19
28.4 Do Not Silently Correct SJ192	19
29 Testing and Acceptance Criteria	19
30 Repository Structure	20
31 Deliverables	20
32 Limitations	21
33 Conclusion	21

1. Executive Summary

This project implements an end-to-end data engineering pipeline for ASG Airlines using flight, booking, passenger, and payment data.

The objective is to transform operational source data into reliable, validated, privacy-aware analytical datasets that can support business intelligence reporting.

The implementation uses Python and Pandas for data ingestion, profiling, cleaning, transformation, validation, and KPI preparation. Power BI is used as the reporting and visualization layer.

The complete logical workflow is:

**Ingestion → Quality Assessment → Cleaning → Transformation → PII Protection
→ Validation → KPI Preparation → Power BI**

The pipeline addresses duplicate records, missing values, inconsistent categorical values, timestamp inconsistencies, invalid temporal records, referential integrity, and protection of passenger-related personally identifiable information.

The resulting Power BI model supports analysis of flight duration, route traffic, airline distribution, booking status, payments, and data-quality anomalies.

2. Problem Statement

ASG Airlines collects operational information through flight operations, booking systems, passenger records, and payment transactions.

The supplied data contains intentional and naturally occurring quality issues such as missing values, duplicate records, inconsistent values, timestamp inconsistencies, and cross-day flight scenarios.

These issues can affect the accuracy of flight-duration calculations, route analysis, airline-level reporting, and operational KPIs.

A standardized data engineering pipeline is therefore required to ingest, clean, validate, transform, and prepare the data for analytical reporting.

The assignment specifically requires appropriate handling of invalid records, missing values, schema inconsistencies, duplicates, standardization, PII protection, business KPIs, Power BI reporting, and detailed documentation.

3. Project Objectives

The main objectives are:

1. Ingest all supplied airline datasets.
2. Understand the structure and quality of the source data.
3. Identify missing, duplicate, malformed, and inconsistent records.
4. Standardize categorical and timestamp fields.

5. Calculate reliable flight durations.
6. Correct legitimate overnight flight scenarios.
7. Detect and handle invalid temporal records.
8. Protect sensitive passenger and booking information.
9. Validate referential integrity.
10. Prepare analytical datasets for aggregation.
11. Calculate required business KPIs.
12. Build an interactive Power BI dashboard.
13. Document architecture, data flow, data model, assumptions, and transformation decisions.

4. Dataset Description

Four source datasets were used.

Table 1: Source datasets

Dataset	Description	Primary Analytical Purpose
Flights	Flight identifier, airline, route, departure and arrival timestamps	Flight operations analysis
Bookings	Booking and passenger-flight relationships	Booking analysis
Passengers	Passenger information	Passenger dimension
Payments	Payment transactions associated with bookings	Financial analysis

4.1. Flight Dataset Structure

The main flight dataset contains:

- `flight_id`
- `airline`
- `source`
- `destination`
- `departure_time`
- `arrival_time`

5. End-to-End Solution Architecture

The implemented architecture uses local Python/Pandas processing. The assignment permits local implementation when cloud resources are unavailable, while Azure is recommended rather than mandatory.

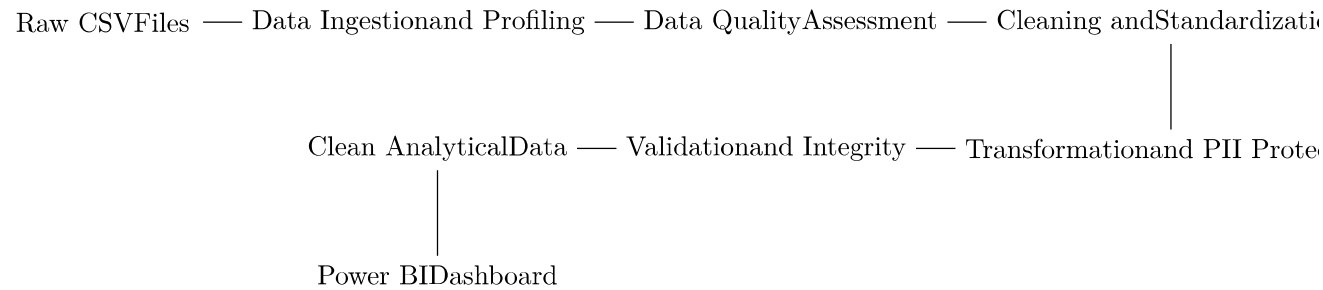


Figure 1: End-to-end solution architecture

5.1. Architecture Layers

Table 2: Architecture layers and responsibilities		
Layer	Technology	Responsibility
Source	CSV	Operational source data
Ingestion	Python / Pandas	Load and inspect source data
Quality	Python / Pandas	Profile missing values, duplicates and invalid records
Transformation	Python / Pandas	Standardization, duration calculation and PII protection
Validation	Python / Pandas	Integrity and business-rule validation
Analytical	Cleaned CSV	Reporting-ready datasets
Reporting	Power BI	Interactive dashboard and KPI analysis

6. Data Flow Diagram

The following DFD describes the movement of information from raw datasets through the transformation pipeline.

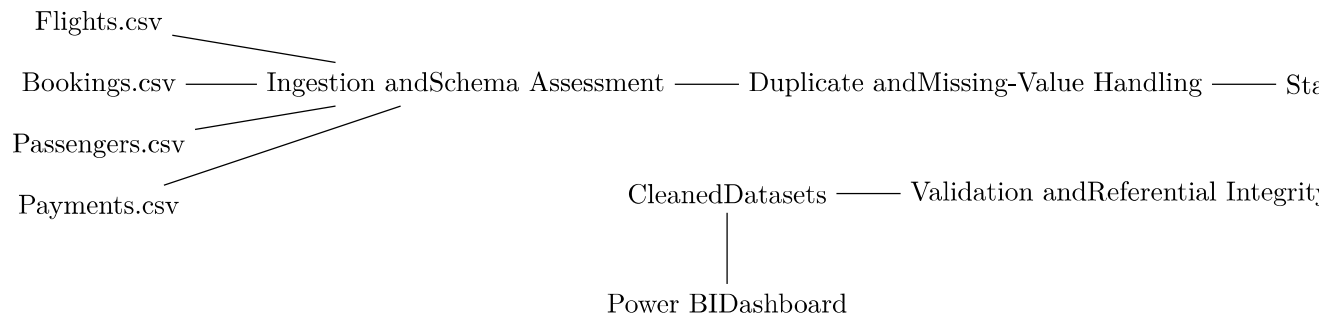


Figure 2: Logical data flow diagram

7. Data Quality Assessment

The source data was assessed before transformation.

The assessment focused on:

- Schema consistency
- Data types
- Missing values
- Exact duplicate rows
- Duplicate business identifiers
- Identifier structure
- Timestamp consistency
- Flight duration validity
- Referential integrity
- Payment-value validity

7.1. Duplicate Analysis

The following duplicate patterns were observed.

Table 3: Duplicate analysis			
Dataset	Exact Duplicates	Business-Key Observation	Decision
Flights	15	One repeated flight identifier with materially different records	Remove exact duplicates only
Bookings	0	No duplicate booking identifiers	No duplicate-row removal
Passengers	0	39 repeated passenger identifiers	Retain records and create unique passenger dimension
Payments	0	363 repeated booking identifiers	Retain because multiple transactions may exist

Exact duplicates are removed because the records contain identical information.

A repeated identifier is not automatically treated as a duplicate record because the records may represent different business events.

7.2. Flight Identifier Validation

The expected flight identifier structure was checked using a pattern-based validation rule.

The expected structure is a five-character identifier consisting of two alphanumeric characters followed by three digits.

The validation pattern was:

$$\sim[A-Z0-9]\{2\}d\{3\}$$

No malformed flight identifiers were identified through this structural validation.

8. Missing-Value Handling

Missing values were handled according to the semantic meaning of each field.

Table 4: Missing-value strategy

Field	Treatment	Reason
Airline	Impute only where a valid flight-prefix mapping exists	Uses evidence already present in the dataset
Booking status	Missing values set to UNKNOWN	Avoids inventing a business status
Payment amount	Retained as missing	Zero would incorrectly represent a payment
Key identifiers	Validated to ensure no unresolved missing keys	Required for relationships

8.1. Airline Imputation Logic

Missing or unknown airline values were resolved using observed flight-prefix mappings.

The mappings identified from valid records were:

Table 5: Observed flight-prefix mappings

Flight Prefix	Airline
6F	IndiGo
AI	Air India
SJ	SpiceJet
UK	Vistara

The flight prefix is therefore used as a lookup key rather than being directly interpreted as the airline name.

9. Data Standardization

Categorical text values were standardized before analysis.

The transformation includes:

- Removing leading and trailing whitespace from airline values.
- Removing whitespace from source values.
- Removing whitespace from destination values.
- Converting source and destination values to uppercase.

- Converting timestamp fields to datetime representation.

For example:

```
source = source.str.strip().str.upper()
destination = destination.str.strip().str.upper()
```

10. Flight Duration Calculation

Flight duration is derived from departure and arrival timestamps.

The basic calculation is:

$$\text{Duration} = \text{Arrival Time} - \text{Departure Time}$$

The duration is stored as a time interval and converted into hours for KPI calculations:

$$\text{Duration Hours} = \frac{\text{Duration in Seconds}}{3600}$$

10.1. Duration Validation

After calculating duration, the pipeline checks whether the resulting duration is valid.

A valid analytical flight record must satisfy:

$$\text{Arrival Timestamp} > \text{Departure Timestamp}$$

after legitimate overnight adjustments have been applied.

11. Overnight Flight Handling

Flights can legitimately cross midnight.

For example, a flight that departs at 22:30 and arrives at 01:30 may have an arrival clock time earlier than the departure clock time.

In such cases, the arrival date is advanced by one day when the source timestamp pattern supports a legitimate overnight interpretation.

Conceptually:

$$\text{Adjusted Arrival} = \text{Arrival} + 1 \text{ day}$$

The adjustment is not applied blindly.

The date component is also checked to distinguish a legitimate overnight flight from a corrupted timestamp.

11.1. Invalid Temporal Record

Flight SJ192 contained an arrival timestamp on the previous calendar day relative to its departure.

The record therefore produced an invalid temporal relationship.

Adding one day to the timestamp would still result in an implausibly long duration and would require an unsupported assumption about the intended source date.

The record was therefore excluded from the analytical flight dataset.

Related dependent booking and payment records were also removed where necessary to maintain referential integrity.

This decision follows a conservative data-engineering principle: when a source value cannot be corrected reliably from available evidence, it should not be silently fabricated.

12. Statistical Anomaly Detection

An IQR-based method was used to identify unusually long flight durations.

The interquartile range is:

$$IQR = Q_3 - Q_1$$

The upper anomaly boundary is:

$$Upper\ Bound = Q_3 + 1.5(IQR)$$

For the cleaned flight data:

$$Q_1 = 1.65 \text{ hours}$$

$$Q_3 = 3.89 \text{ hours}$$

Therefore:

$$IQR = 3.89 - 1.65$$

$$IQR = 2.24 \text{ hours}$$

The upper boundary is:

$$3.89 + (1.5 \times 2.24)$$

$$\approx 7.24 \text{ hours}$$

No remaining flight records exceeded this statistical threshold.

13. PII Protection

The passenger and booking datasets contain personally identifiable and sensitive information.

The assignment requires passenger PII to be protected through masking, hashing, encryption, or an equivalent approach.

The implemented protection strategy is shown below.

Table 6: PII protection strategy

Field	Protection	Reason
Aadhaar	SHA-256 hashing	Protects the original sensitive identifier
Passport number	SHA-256 hashing	Protects the original passport identifier
Email	Masking	Reduces direct identification
Phone	Masking with last four digits retained	Maintains limited analytical reference
Date of birth	Generalized to birth year	Reduces precision of personal information
Emergency contact name	Removed	Not required for analytical KPIs
Emergency contact phone	Removed	Not required for analytical KPIs

13.1. Hashing

Sensitive identifiers such as Aadhaar and passport numbers are protected using SHA-256 hashing.

Conceptually:

$$H(x) = \text{SHA256}(x)$$

The resulting hash is stored instead of the original identifier.

13.2. Email Masking

Email addresses are transformed into a reduced-identification representation.

For example:

`original@example.com`

can be represented in masked form while retaining the domain.

13.3. Phone Masking

Phone numbers are masked while retaining only the final four digits for limited operational reference.

13.4. Data Minimization

Fields that are not necessary for analytical reporting are removed from the analytical dataset. This follows the principle of exposing only the information required for the intended reporting purpose.

14. Privacy and Access Control

The analytical repository should not contain raw PII.

The production architecture can separate data into restricted and analytical zones.



Figure 3: Logical privacy and access-control model

In a production environment, raw PII should be accessible only to authorized users and processing services.

The reporting layer should expose only protected or aggregated information.

Enterprise identity management and role-based access control are considered deployment requirements rather than part of the local notebook implementation.

15. Data Validation

Validation is performed after transformation to ensure that the cleaned datasets remain analytically consistent.

Table 7: Validation results

Validation Check	Result
Exact duplicates after cleaning	0
Missing key identifiers	0
Invalid flight durations	0
Bookings without matching flights	0
Bookings without matching passengers	0
Payments without matching bookings	0
Negative payment amounts	0
Missing payment amounts	78
Statistical duration anomalies	0

The 78 missing payment values are not treated as validation failures because the business rule is to preserve them as missing rather than incorrectly replacing them with zero.

16. Analytical Data Model

The Power BI model uses dedicated dimensions to avoid ambiguous relationships and incorrect aggregation.

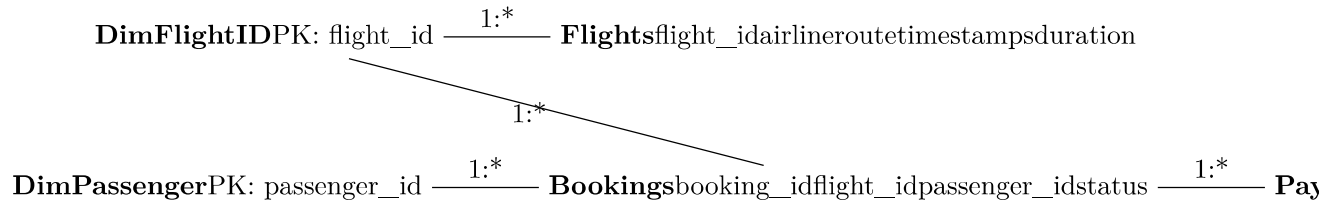


Figure 4: Power BI analytical data model

16.1. Relationship Design

The model contains four active one-to-many relationships:

1. DimFlightID → Flights
2. DimFlightID → Bookings
3. DimPassenger → Bookings
4. Bookings → Payments

All relationships use single-direction filtering.

A direct Flights-to-Bookings relationship is intentionally avoided because the model already provides a shared flight dimension.

17. Duplicate-Key Modelling Considerations

The source data contains repeated passenger identifiers and repeated payment booking identifiers.

Passenger records are therefore not directly used as a one-side unique relationship table.

A unique **DimPassenger** table is created for the Power BI model.

Similarly, payment records remain at their transaction grain.

This prevents a join from multiplying payment values when multiple payment records exist for a booking.

18. Business KPIs

The assignment requires the following analytical metrics:

- Average Flight Duration
- Route-wise Traffic

- Delays / Anomalies
- Distribution of Flights by Airline

Additional KPIs were also included.

Table 8: Final KPI summary

KPI	Definition	Result
Total Flights	Count of cleaned flight records	1,004
Average Flight Duration	Average validated flight duration	2.74 hours
Total Bookings	Count of cleaned booking records	999
Cancelled Bookings	Count of CANCELLED bookings	314
Cancellation Rate	Cancelled bookings divided by total bookings	31.43%
Total Payment Amount	Sum of valid payment amounts	INR 7,370,442.88
Statistical Duration Anomalies	Durations above IQR threshold	0

19. Airline Distribution

The cleaned flight data contains the following airline distribution.

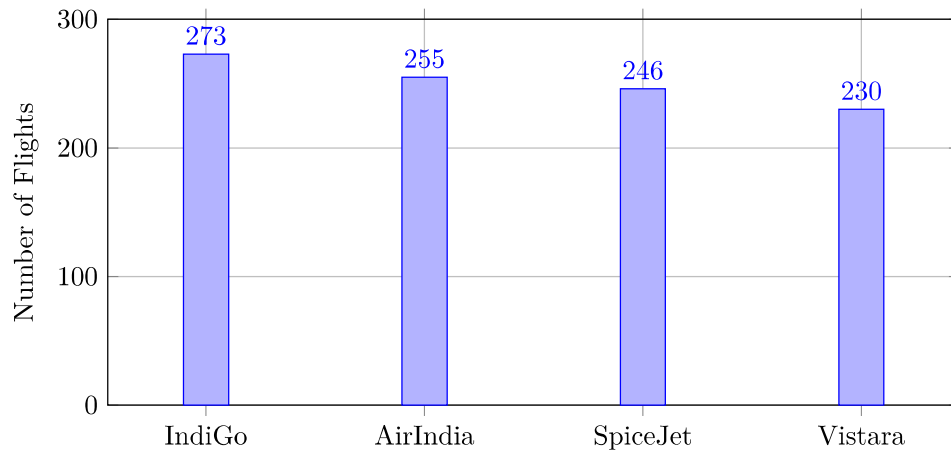


Figure 5: Distribution of flights by airline

20. Route-wise Traffic

The highest-volume routes in the cleaned flight data are:

Table 9: Top routes by traffic

Route	Flights
BOM to CCU	90
CCU to DEL	72
MAA to BLR	65
BLR to BOM	60
HYD to MAA	57

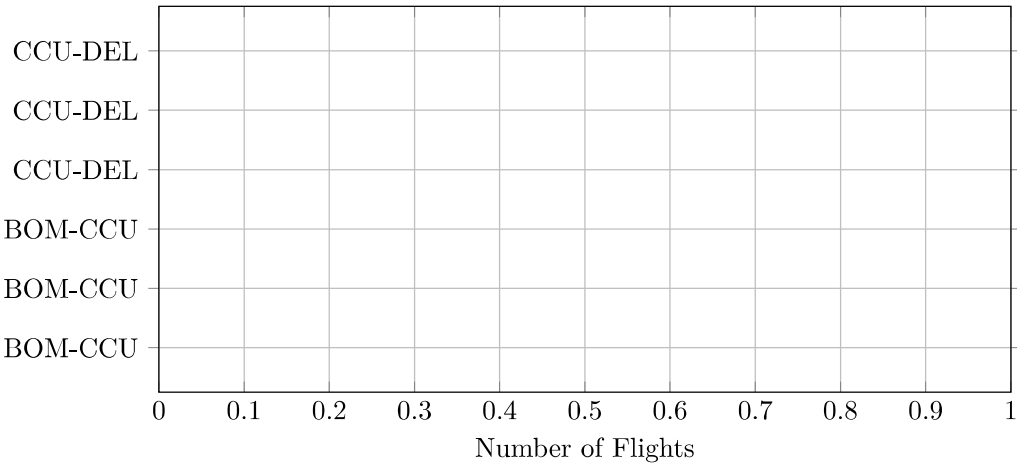


Figure 6: Top routes by flight traffic

21. Booking Status Analysis

The final booking-status distribution is:

Table 10: Booking status distribution

Status	Bookings
CONFIRMED	319
CANCELLED	314
PENDING	291
UNKNOWN	45
INVALID	30

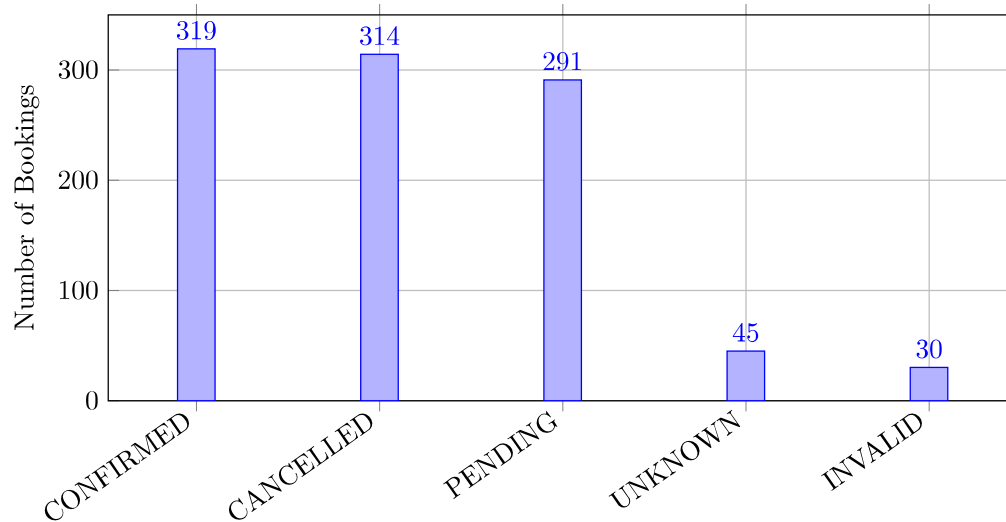


Figure 7: Booking status distribution

22. Delay and Anomaly Analysis

A traditional flight-delay metric could not be calculated because the supplied data does not contain scheduled departure and scheduled arrival timestamps.

Actual delay would require:

$$\text{Delay} = \text{Actual Departure} - \text{Scheduled Departure}$$

or an equivalent scheduled-versus-actual comparison.

Since scheduled timestamps are unavailable, creating a delay KPI would require inventing missing information.

Instead, the project reports:

- Invalid temporal records
- Flight-duration validation
- Statistical duration anomalies

One invalid temporal record was removed and no remaining flight was identified as an IQR duration anomaly.

23. Power BI Dashboard

The Power BI dashboard provides an interactive view of operational flight and booking performance.

The dashboard contains KPI cards and analytical visuals.

23.1. KPI Cards

The main KPI cards include:

- Total Flights
- Average Flight Duration
- Total Bookings
- Total Payment Amount
- Cancellation Rate
- Average Payment Amount
- Statistical Duration Anomalies

23.2. Interactive Visuals

The dashboard includes:

- Flights by Airline
- Average Flight Duration by Airline
- Booking Status Distribution
- Top Routes by Flight Traffic
- Flight Volume by Date
- Airline slicer

23.3. Operational Interpretation

The dashboard allows users to answer questions such as:

- Which airline operates the largest number of flights?
- Which routes have the highest traffic?
- What is the average flight duration?
- What proportion of bookings are cancelled?
- How are booking statuses distributed?
- Are there statistically unusual flight durations?

24. Data Quality Governance

The pipeline follows several governance principles:

- Validate data before analytical use.
- Preserve missing values when imputation would create misleading information.
- Do not silently modify questionable source records.
- Protect sensitive identifiers.
- Minimize unnecessary personal information.
- Maintain referential integrity between related datasets.
- Keep transformation decisions documented.

25. Scalability and Performance

The current dataset is sufficiently small for local Python/Pandas processing.

For larger datasets, the logical architecture can be migrated to cloud or distributed technologies.

Table 11: Potential production-scale implementation

Current Approach	Potential Scaled Approach
CSV files	Cloud object storage / data lake
Python / Pandas	PySpark / Databricks / Fabric
Notebook execution	Scheduled orchestration
CSV analytical output	Warehouse or lakehouse tables
Local validation	Centralized data-quality framework
Power BI	Governed Power BI semantic model

Performance improvements for larger volumes could include:

- Incremental ingestion
- Partitioning by operational date
- Early filtering
- Distributed transformation
- Column pruning
- Optimized joins
- Aggregation before reporting where appropriate

26. Error Handling and Logging

The pipeline uses validation outputs to identify data-quality problems during processing.

Important failure categories include:

- Missing input files
- Unexpected columns
- Invalid timestamp values
- Missing key identifiers
- Invalid payment values
- Referential-integrity failures

For a production implementation, these validation events should be written to centralized logs with timestamps, pipeline run IDs, and error categories.

27. Assumptions and Business Rules

The following assumptions were applied:

1. Exact duplicate rows are treated as redundant and removed.
2. Repeated identifiers are not automatically treated as duplicate business events.
3. Repeated passenger identifiers are handled through a unique passenger dimension.
4. Multiple payment records for a booking are retained.
5. Missing booking status is represented as `UNKNOWN`.
6. Missing payment amounts remain missing.
7. Zero is not used as a replacement for a missing payment.
8. Airline imputation is performed only when a valid observed mapping exists.
9. Legitimate overnight flights are allowed to cross midnight.
10. An invalid timestamp is not corrected by inventing an unsupported date.
11. Flight `SJ192` is excluded because its temporal information cannot be reliably corrected.
12. Traditional delay cannot be calculated without scheduled timestamps.
13. Statistical duration anomalies are identified using the IQR method.
14. Raw PII should not be published in the repository.

28. Key Design Decisions

Several design decisions were made to prioritize analytical correctness.

28.1. Do Not Treat Every Repeated ID as a Duplicate

A business identifier can occur more than once without representing a duplicate row.

Therefore, exact duplicate detection and business-key analysis are performed separately.

28.2. Do Not Convert Missing Payments to Zero

A missing payment amount does not necessarily mean that the payment was zero.

Therefore, the missing value is retained and represented using an amount-status field.

28.3. Do Not Invent Scheduled Flight Times

Delay analysis requires scheduled timestamps.

Since those timestamps are not supplied, a traditional delay metric is not fabricated.

28.4. Do Not Silently Correct SJ192

The temporal inconsistency cannot be resolved from available evidence.

Removing the record from analytical reporting is therefore more defensible than manufacturing a corrected timestamp.

29. Testing and Acceptance Criteria

The following checks are used as acceptance criteria:

- All required source datasets load successfully.
- Required columns are present.
- Timestamp fields can be converted to datetime.
- Exact duplicate rows are removed.
- Key identifiers contain no unresolved missing values.
- Flight durations are valid.
- Legitimate overnight cases are handled.
- Invalid temporal records are identified.
- Referential integrity is maintained.
- Negative payment amounts are absent.
- PII fields are protected.
- KPI calculations reconcile with cleaned datasets.
- Power BI relationships remain one-to-many.
- Dashboard filtering works correctly.

30. Repository Structure

The final repository is organized around the pipeline, analytical outputs, reporting assets, and documentation.

```

ASG-AIRLINES-ANALYTICS-PIPELINE/
|
+-- README.md
|
+-- ASSIGNMENT_NEOSTAT.ipynb
|
+-- bookings_clean.csv
+-- flights_clean.csv
+-- kpi_summary.csv
+-- passengers_clean.csv
+-- payments_clean.csv
|
+-- powerbi/
|   |
|   +-- ASG_Airlines_Dashboard.pbix
|   +-- dashboard_screenshot.png
|
+-- docs/
|   |
|   +-- ASG_Airlines_Documentation.pdf
|   +-- architecture.png
|   +-- data_flow_diagram.png
|   +-- data_model.png

```

31. Deliverables

The completed project provides the following deliverables:

Table 12: Project deliverables

Deliverable	Description
Pipeline Notebook	Complete ingestion, cleaning, transformation, validation and KPI workflow
Cleaned Dataset	Analytical datasets prepared for reporting
Power BI Report	Interactive operational dashboard
Dashboard Screenshot	Static representation of the completed dashboard
Architecture	End-to-end solution architecture
Data Flow Diagram	Movement of data through the pipeline
Data Model	Analytical relationship model
Documentation	Cleaning, transformation, assumptions and KPI explanations